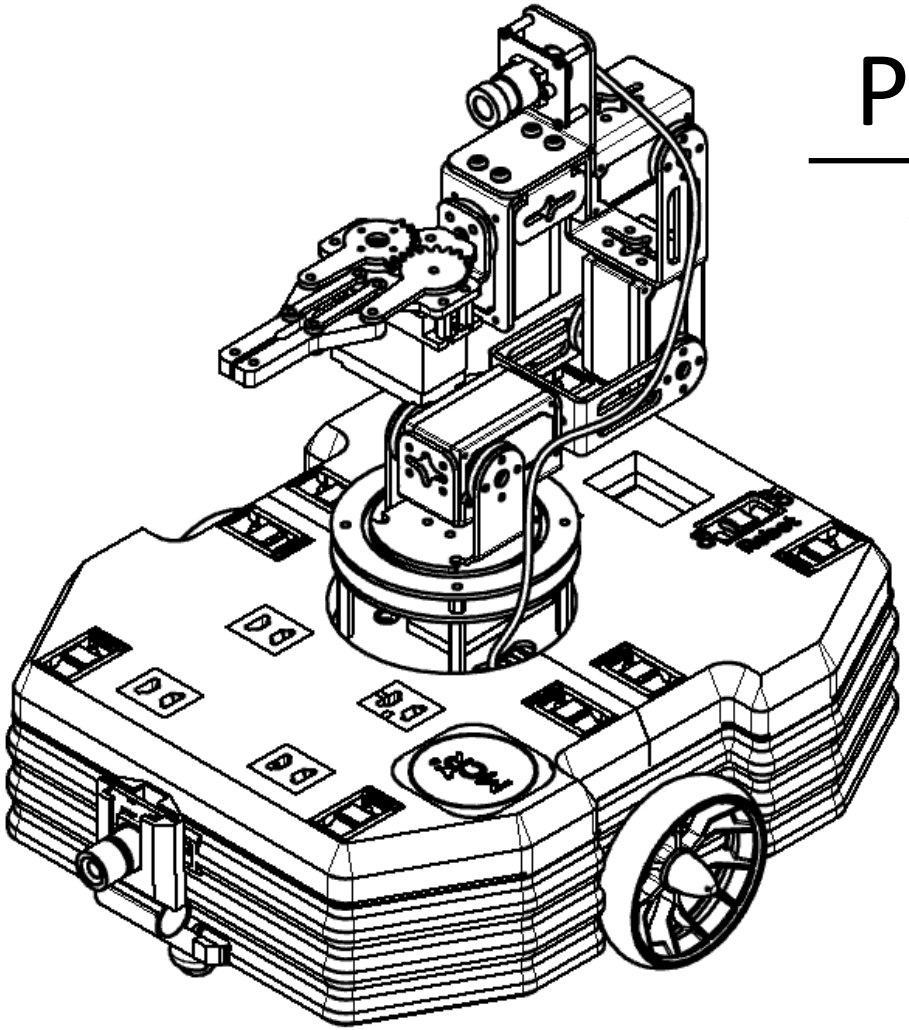
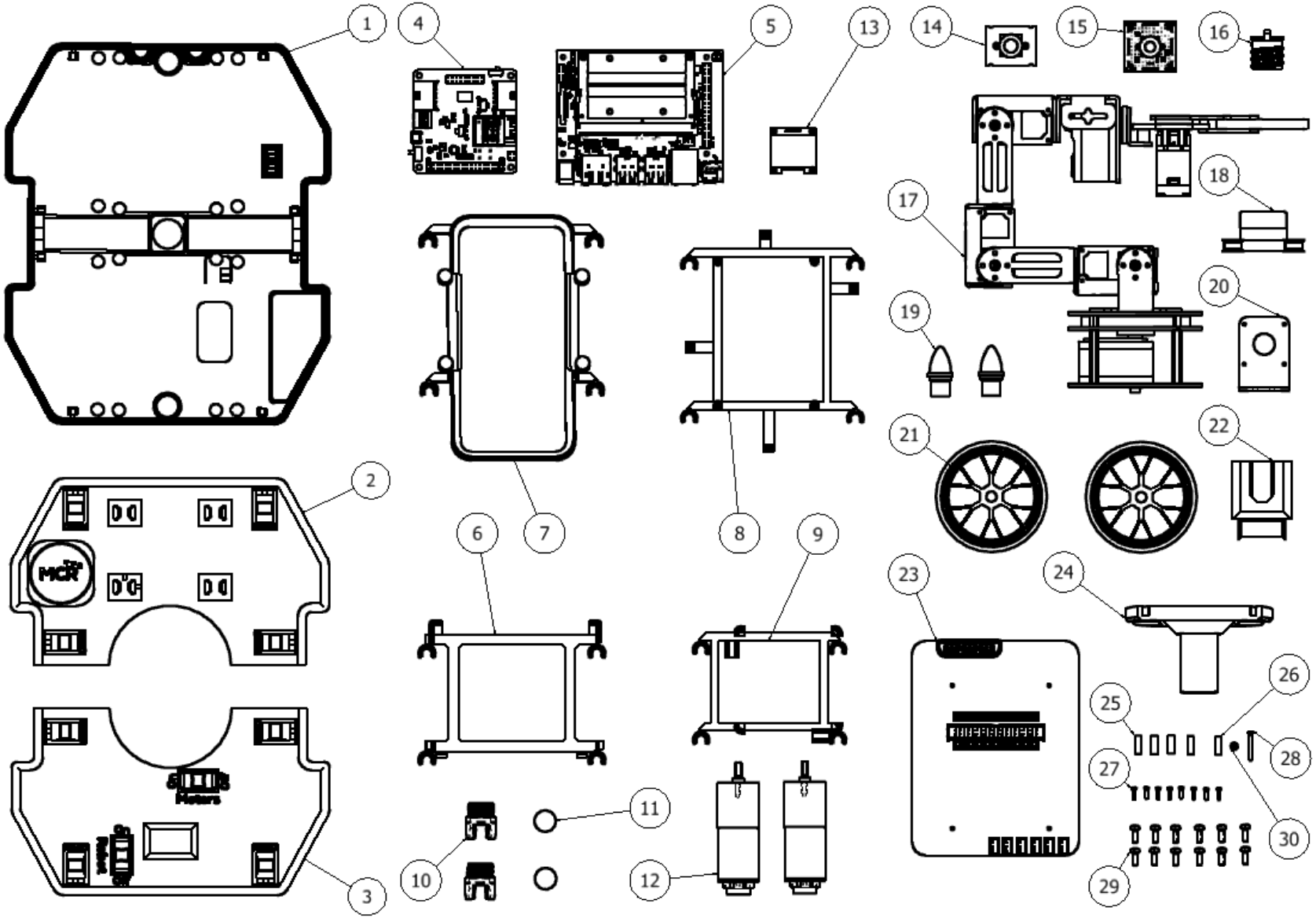
PuzzleArm

Assembly Manual

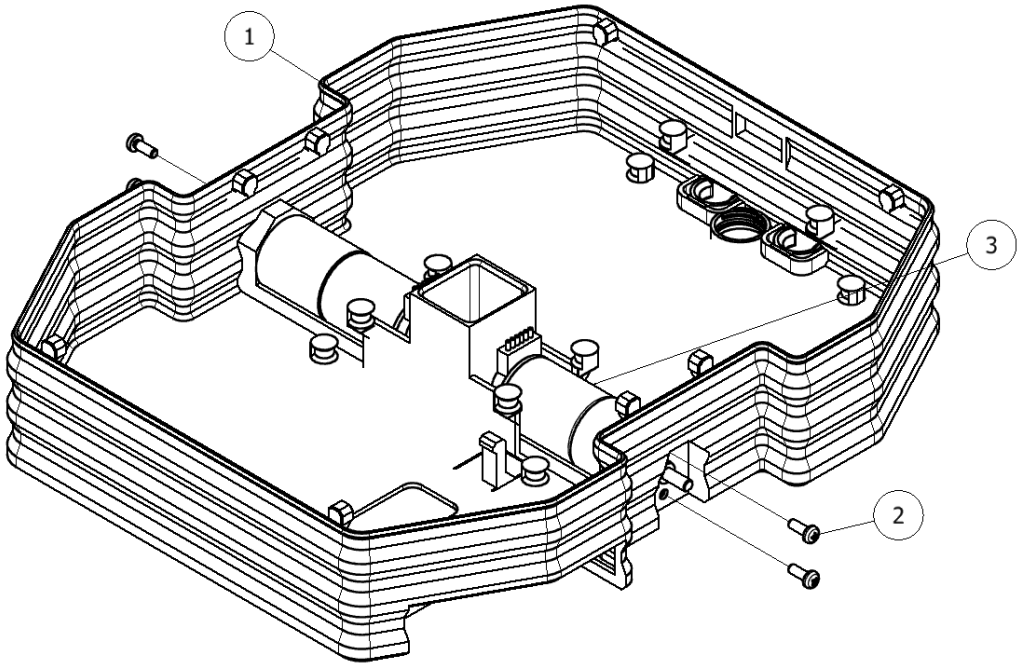


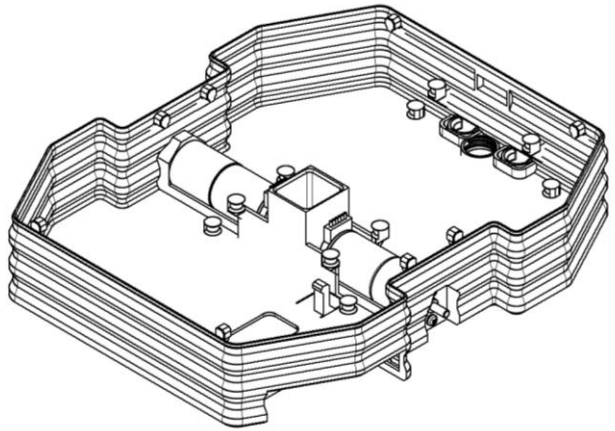
PuzzleArm Components



PARTS LIST		
ITEM	QTY	DESCRIPTION
1	1	Bottom Case
2	1	Front Top Cover
3	1	Rear Top Cover
4	1	MCR2_1200 Control Module
5	1	NVIDIA Jetson Nano™ 4GB Devkit
6	1	Jetson Nano Frame
7	1	Battery Frame
8	1	PCB Frame
9	1	Hackerboard Frame
10	2	Caster Ball Holder
11	2	Caster Ball
12	2	25D DC Motor
13	1	0.96" OLED Display
14	1	USB Arm Camera Module
15	1	USB Camera Module
16	1	Plastic Thumb Screw
17	1	Yahboom Robotic Arm
18	1	Camera Holder Mount
19	2	Bullet Coupler
20	1	Arm Camera Bracket
21	2	Wheel 60 mm diam.
22	1	Camera Holder
23	1	PCB
24	1	Arm Bracket
25	4	Round Metal Spacer
26	1	11 mm Round Nylon Spacer
27	8	M1.6x6 mm Pozi Pan Head Screw
28	1	M2x16 mm Nylon Pozi Pan Head Screw
29	12	M3x8 mm Pozi Pan Head Screw
30	1	M2 Hex Nylon Nut

PARTS LIST		
ITEM	QTY	DESCRIPTION
1	1	Bottom Case
2	4	M3x8 mm Pozi Pan Head Screw
3	2	25D DC Motor





Motor Assembly

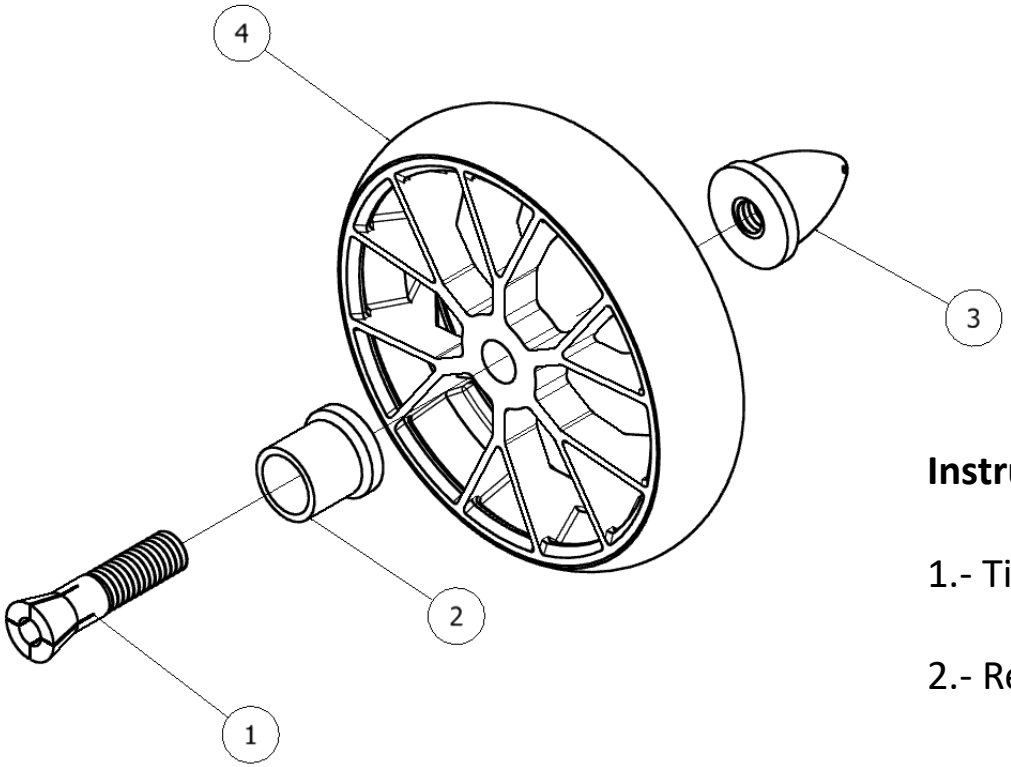
Instructions:

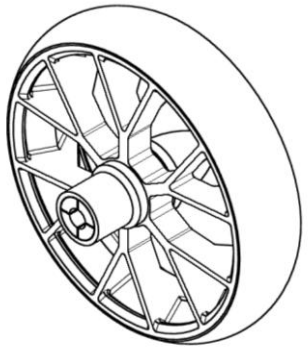
1.- Use one hand to hold the motors and use a screwdriver to properly tighten the screws as shown.

Figure 18 shows the motor connections to the Hackerboard.

<u>Warning:</u> Do not use excessive force.

PARTS LIST		
ITEM	QTY	DESCRIPTION
1	2	Bullet Coupler
2	2	Bullet Coupler Cap
3	2	Bullet Coupler Nut
4	2	Wheel 60 mm. diam.





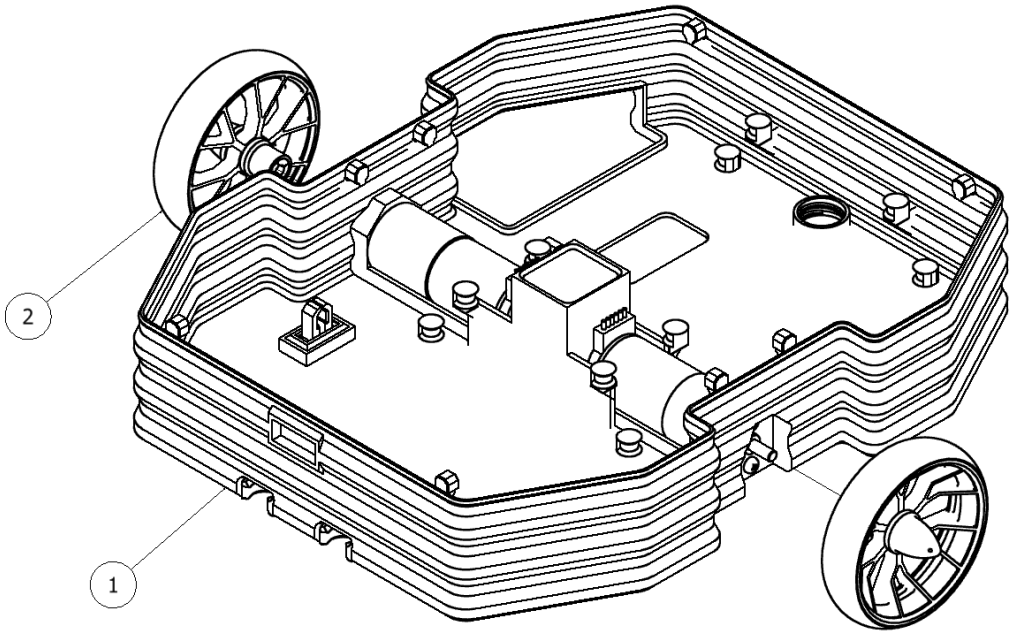
Wheel Coupler Assembly

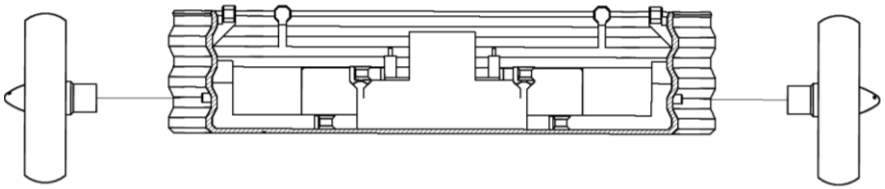
Instructions:

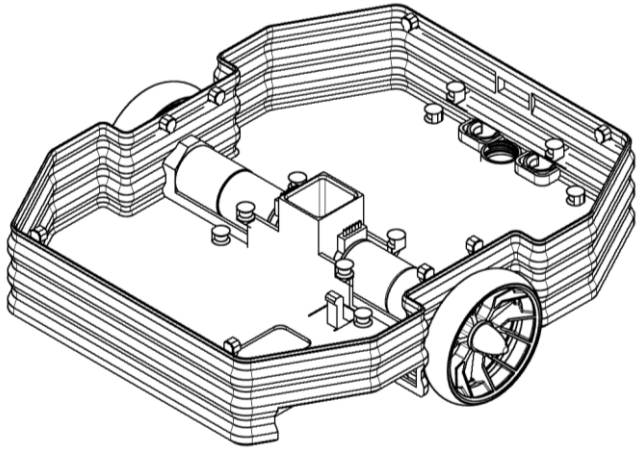
1.- Tighten the bullet coupler as shown.

2.- Repeat the process for the other wheel.

PARTS LIST	
ITEM	DESCRIPTION
1	Motor Assembly
2	Wheel Coupler Assembly







Wheel Base Assembly

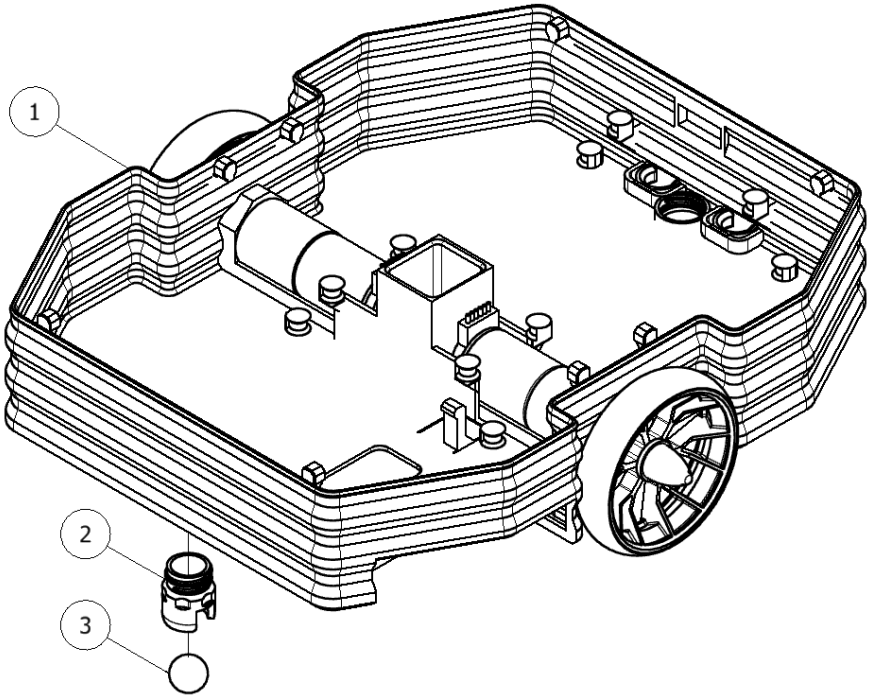
Instructions:

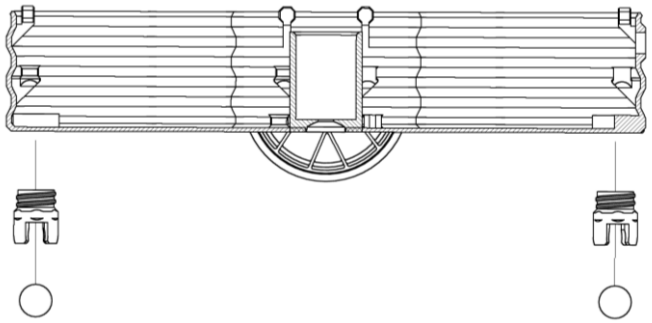
1.- Insert the Wheel Coupler Assembly into the motor shaft.

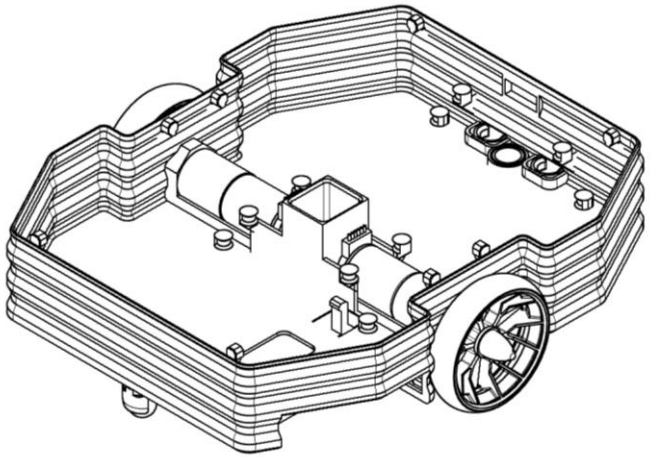
2.- Tighten the motor coupler by hand to the motor shaft and fix the wheels to the motors.

Warning: Do not use excessive force.

PARTS LIST	
ITEM	DESCRIPTION
1	Wheel Base Assembly
2	Caster Ball Holder
3	Caster Ball







Caster Ball Assembly

Instructions:

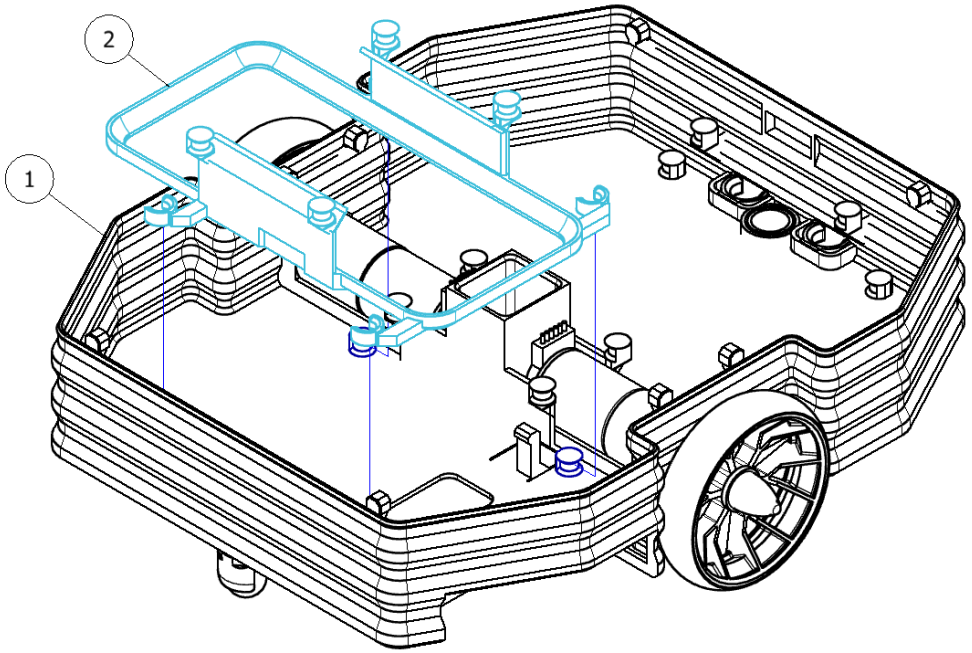
1.- Align and screw the caster ball holder into place until tight.

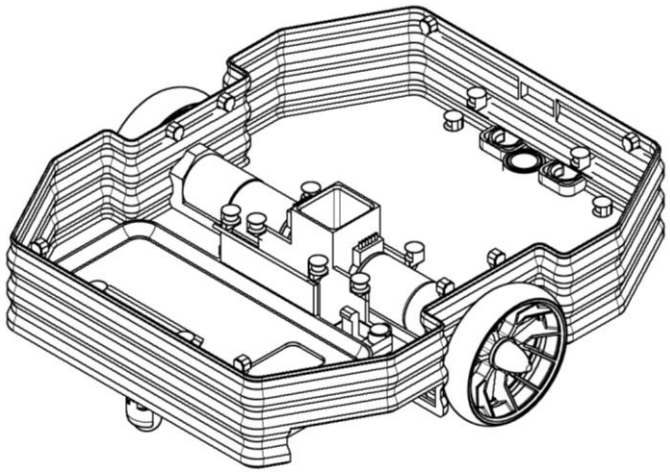
2.- Manually press the caster ball (metal ball) into the holder.

3.- Repeat the process for the other caster ball.

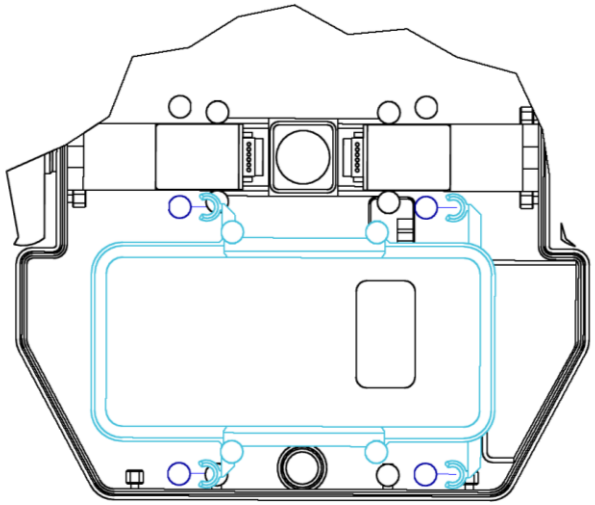
Warning: Do not use excessive force.

PARTS LIST	
ITEM	DESCRIPTION
1	Caster Ball Assembly
2	Battery Frame





Battery Frame Assembly



Instructions:

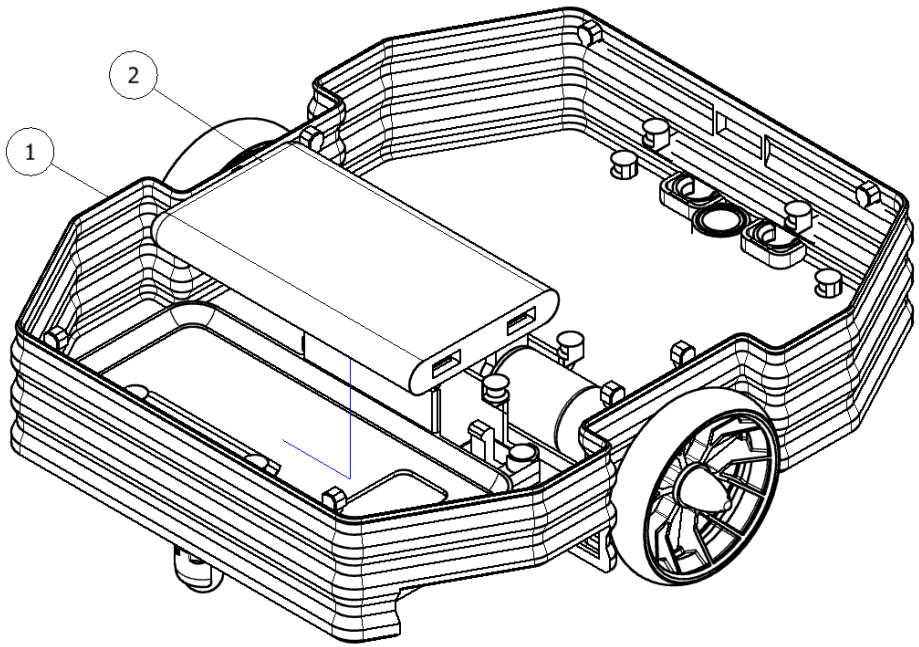
1.- Locate the mounting posts indicated in the diagram.

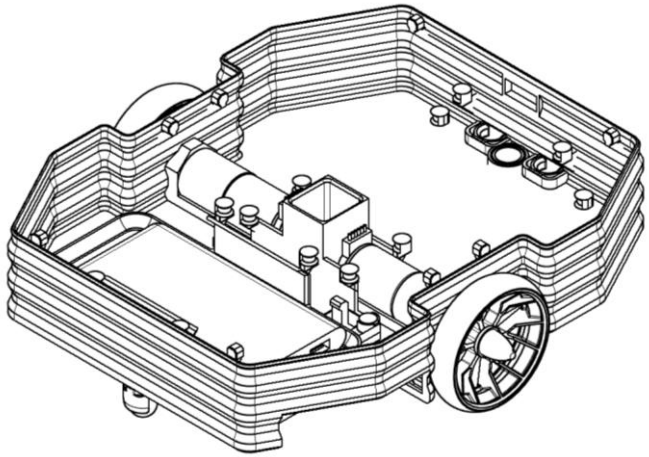
2.- Align the frame with the mounting posts.

3.- Slide the frame forward until it clicks firmly into place.

Warning: Do not force the component. Verify that all four tabs are properly seated in the mounting posts.

PARTS LIST	
ITEM	DESCRIPTION
1	Battery Frame Assembly
2	Powerbank





Battery Mount

Instructions:

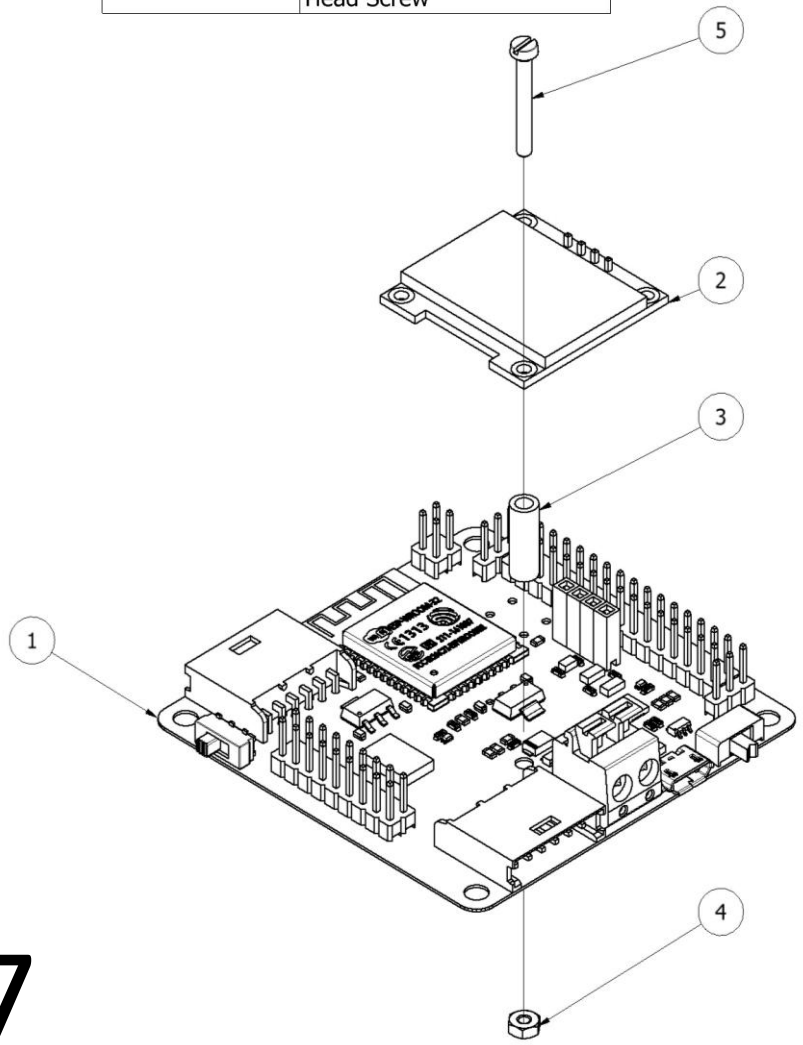
1.- Orient the powerbank with the USB ports facing toward the opening in the base.

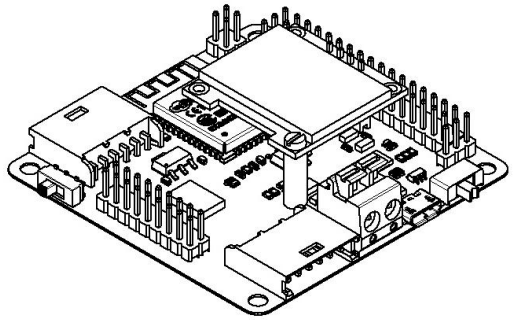
2.- Lower the power bank vertically into the battery frame.

3.- Slide it backward against the frame wall until it is fully seated. **Note:** The powerbank rests loosely within the compartment and is not mechanically locked in place.

Recommended Battery:
Powerbank, 10,000 mAh
5 V, 3 A (15 W). Powerbank Not Included.

PARTS LIST	
ITEM	DESCRIPTION
1	MCR2_1200 Puzzle-Bot Control Module
2	0.96" OLED Display
3	11 mm Round Nylon Spacer
4	M2 Hex Nylon Nut
5	M2x16 mm Nylon Pozi Pan Head Screw





Hackerboard Assembly

Instructions:

1.- Insert the screen headers into the Hackerboard female headers.

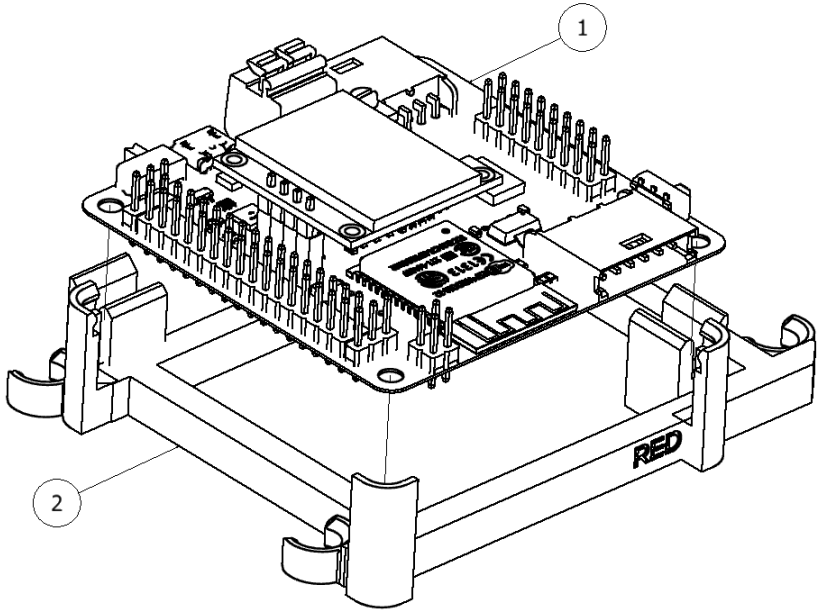
2.- Insert the screw inside the Nylon spacer as shown in the figure.

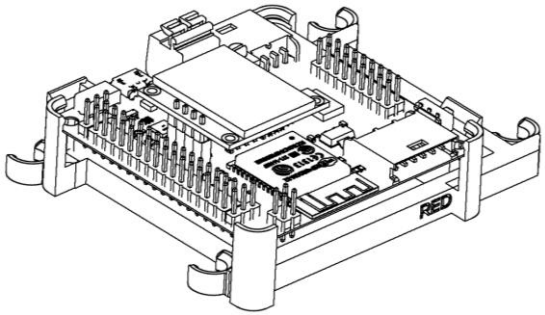
3.- Use a screwdriver to properly tighten the screw and the nut to the Hackerboard.

Figure 17 provides more details of the Hackerboard.

<u>Warning:</u> Do not use excessive force.

PARTS LIST	
ITEM	DESCRIPTION
1	Hackerboard Assembly
2	Hackerboard Frame





Hackerboard Frame

Instructions:

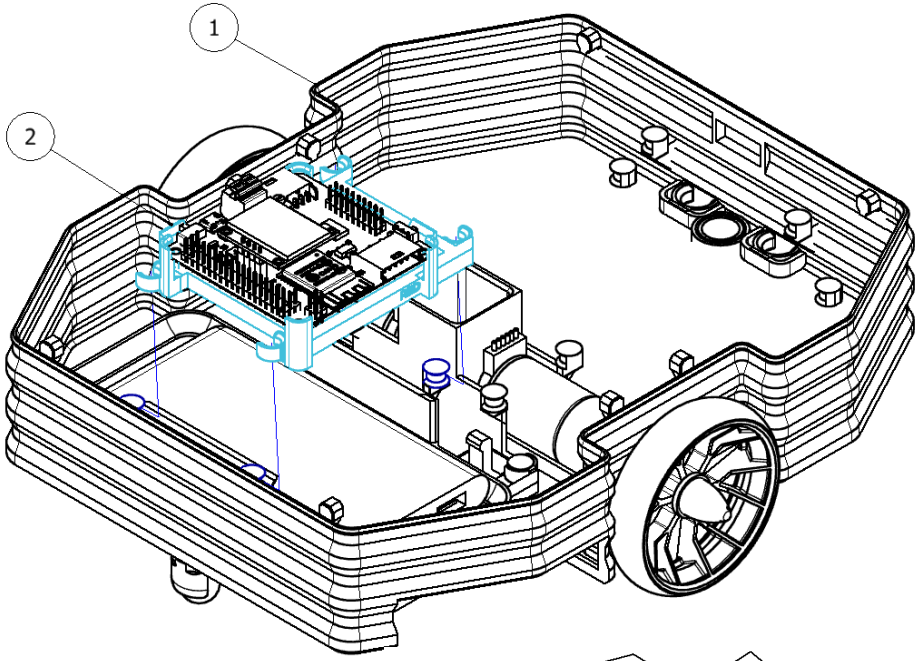
1.- Verify the board's orientation.

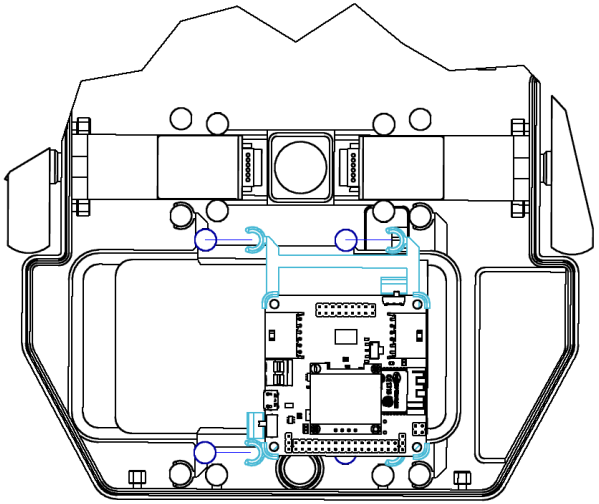
2.- Align the board with the snap-fit retention clips of the Hackerboard Frame.

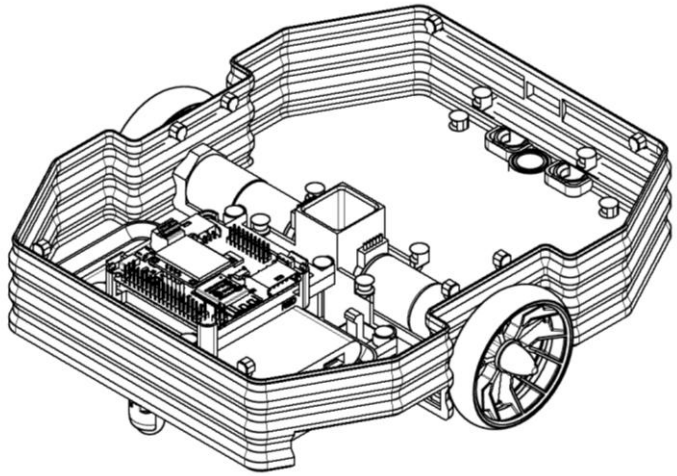
3.- Gently press down on the outer corners of the board until each clip clicks firmly into place.

Warning: Press only along the bare Hackerboard edges. Do not apply force to pins, the screen, or electronic components to prevent damage.

PARTS LIST	
ITEM	DESCRIPTION
1	Battery Mount
2	Hackerboard Frame







Hackerboard Frame Assembly

Instructions:

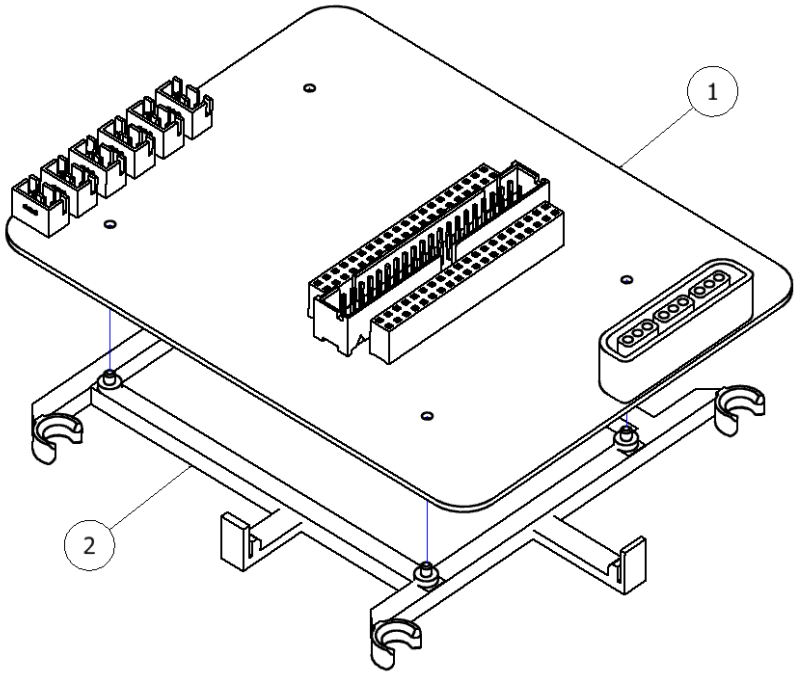
1.- Locate the mounting posts indicated in the diagram.

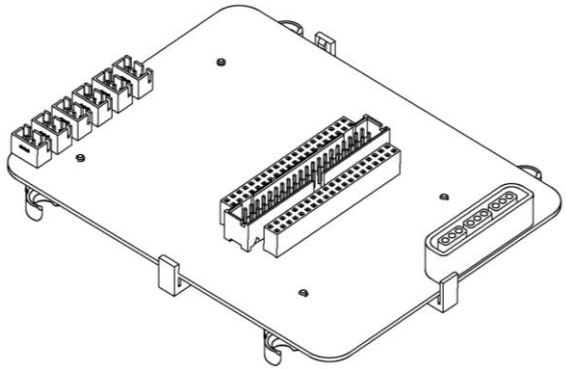
2.- Align the frame with the mounting posts.

3.- Slide the frame forward until it clicks firmly into place.

Warning: Do not force the component. Verify that all four tabs are properly seated in the mounting posts.

PARTS LIST	
ITEM	DESCRIPTION
1	PCB
2	PCB Frame





PCB Frame

Instructions:

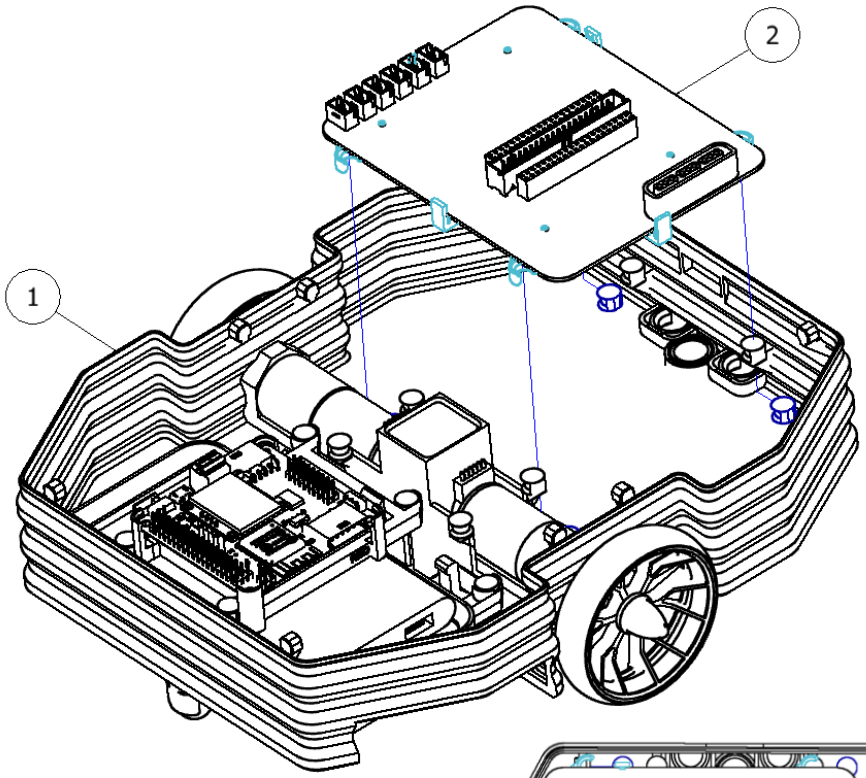
1.- Verify the board's orientation.

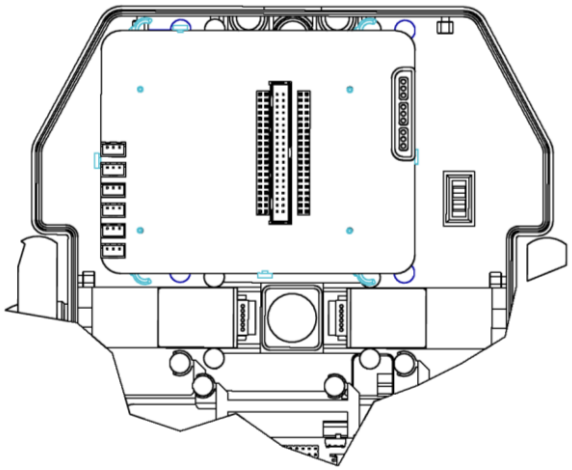
2.- Align the board with the snap-fit retention clips of the PCB Frame.

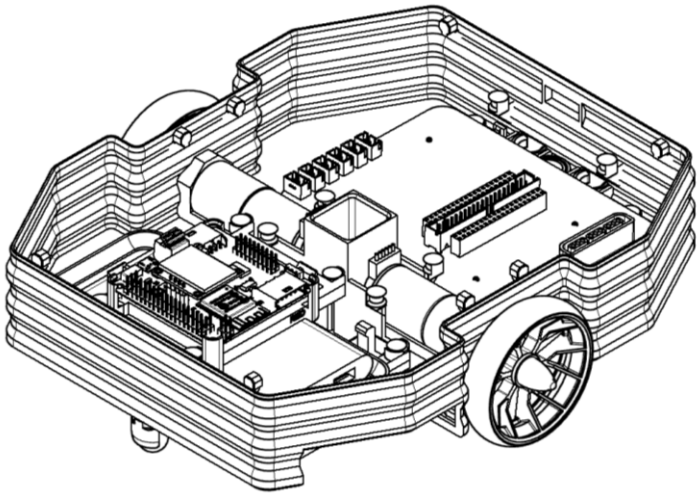
3.- Gently press down on the outer corners of the board until each clip clicks firmly into place.

Warning: Press only along the bare PCB edges. Do not apply force to pins, the screen, or electronic components to prevent damage.

PARTS LIST	
ITEM	DESCRIPTION
1	Hackerboard Frame Assembly
2	PCB Frame







PCB Frame Assembly

Instructions:

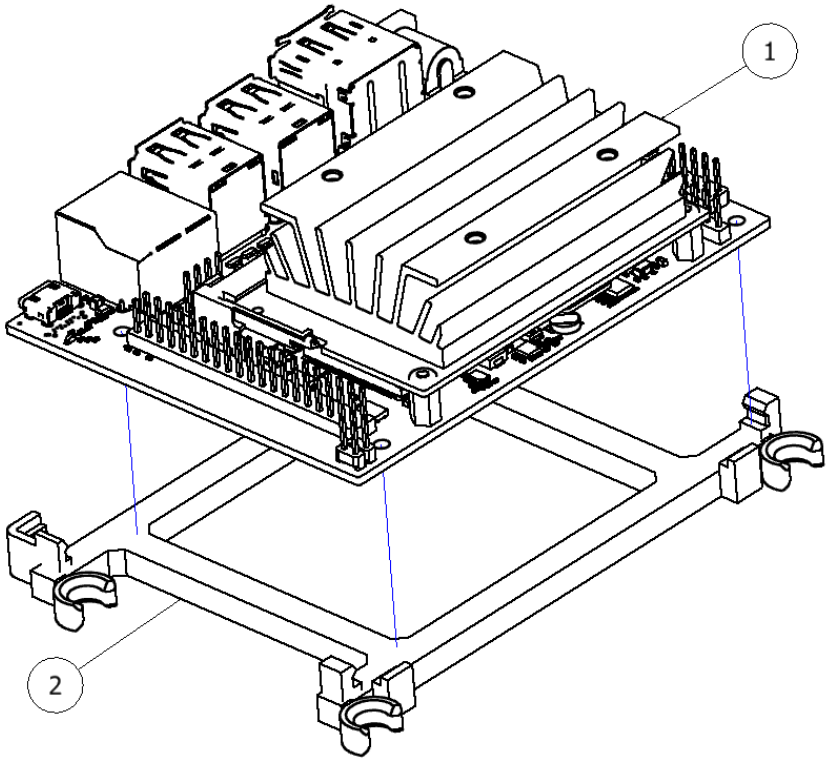
1.- Locate the mounting posts indicated in the diagram.

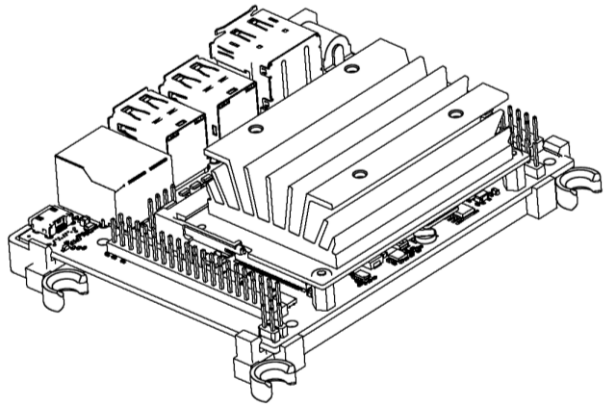
2.- Align the frame with the mounting posts.

3.- Slide the frame forward until it clicks firmly into place.

<u>Warning:</u> Do not force the component. Verify that all four tabs are properly seated in the mounting posts.

PARTS LIST	
ITEM	DESCRIPTION
1	NVIDIA Jetson Nano™ 4GB Devkit
2	Jetson Nano Frame





Jetson Nano Frame

Instructions:

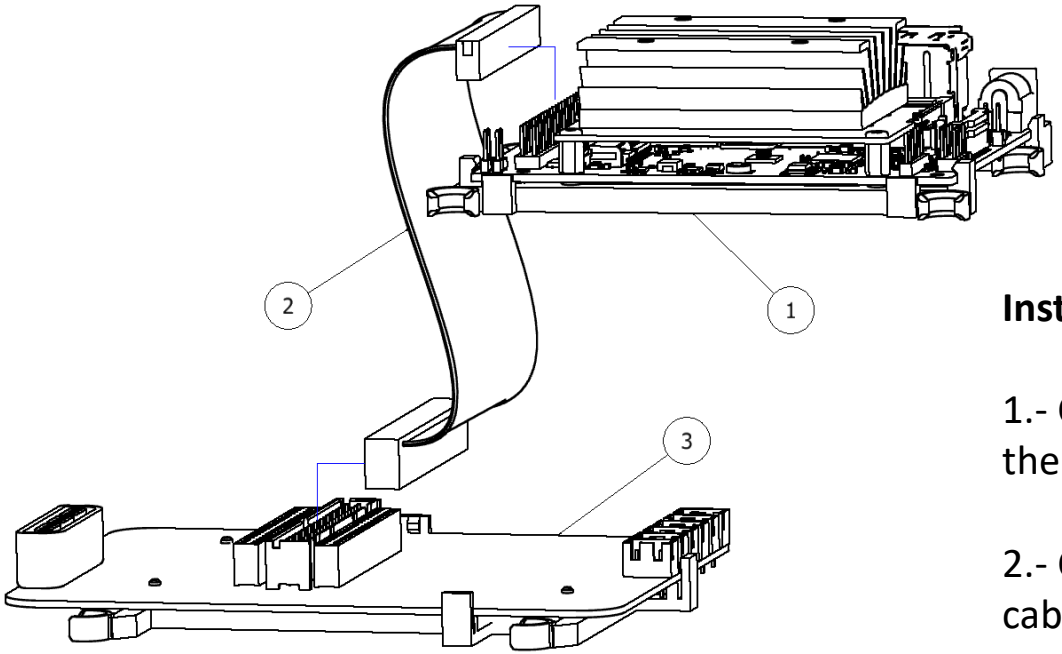
1.- Verify the board's orientation.

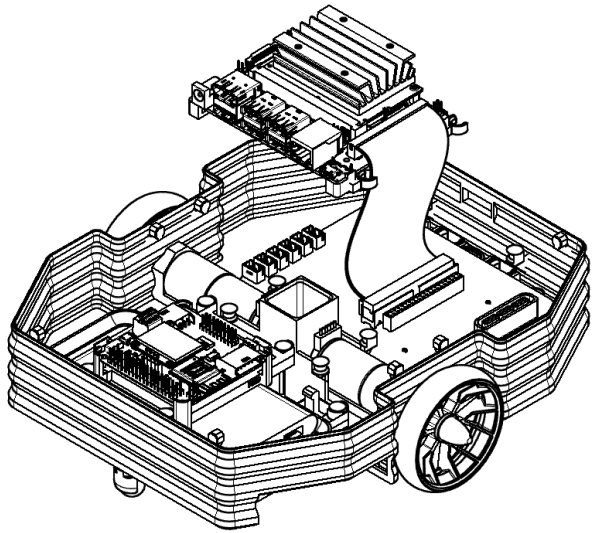
2.- Align the board with the snap-fit retention clips of the Jetson Nano Frame.

3.- Gently press down on the outer corners of the board until each clip clicks firmly into place.

Warning: Press only along the bare Jetson Nano edges. Do not apply force to pins, the screen, or electronic components to prevent damage.

PARTS LIST	
ITEM	DESCRIPTION
1	Jetson Nano Frame
2	Ribbon Cable 40 pins
3	PCB Frame Assembly





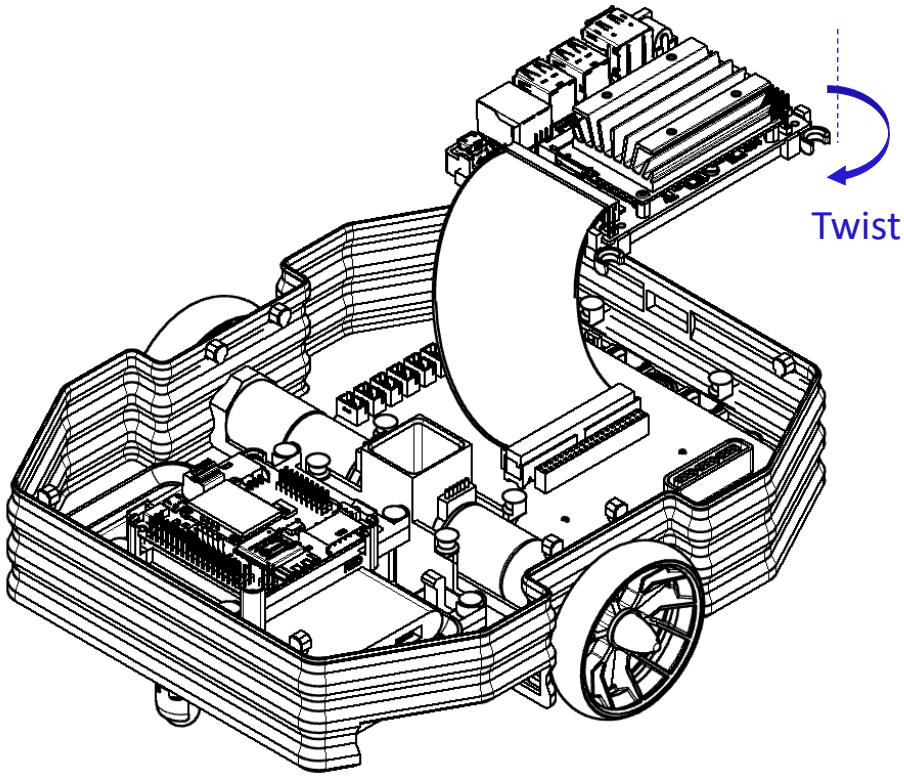
Jetson Nano Flat Cable

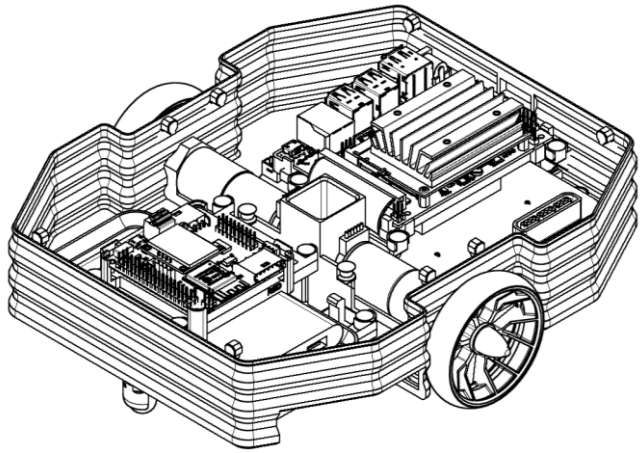
Instructions:

1.- Connect one end of the ribbon cable to the header pins on the Jetson Nano.

2.- Connect the opposite end of the ribbon cable to the middle header on the PCB.

Warning: Pay attention to pin orientation and cable alignment before pressing down to prevent bending pins.





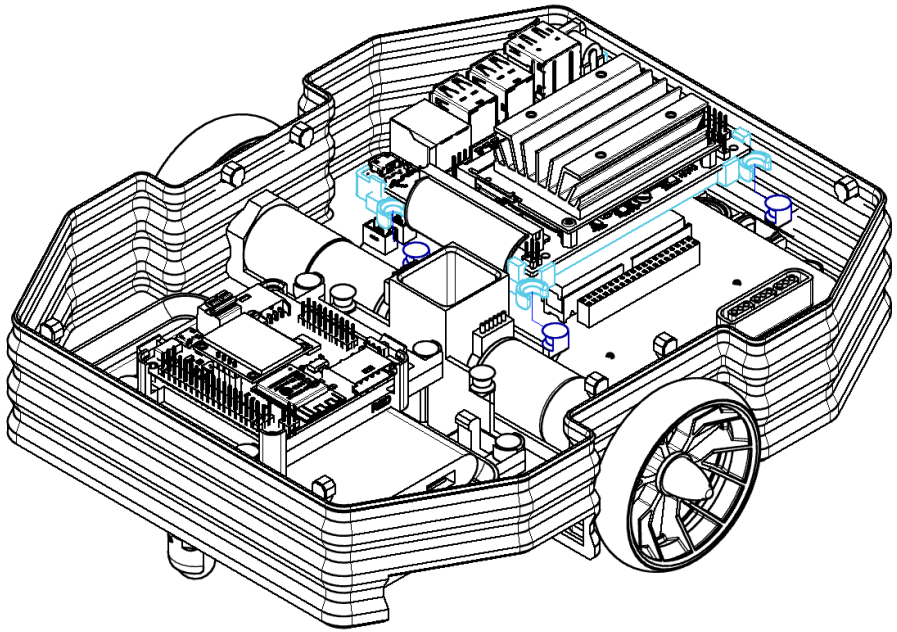
Jetson Nano Frame Assembly

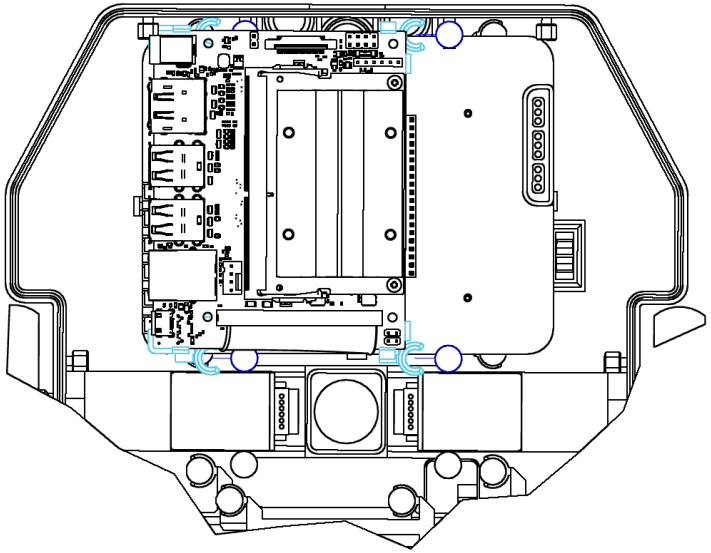
Instructions:

1.- Gently twist the Jetson Nano Frame 90° clockwise, allowing the ribbon cable to fold naturally without pinching.

Warning: Do not pull or over-twist the ribbon cable to avoid damaging the connectors or pins.

(Continued on next page)





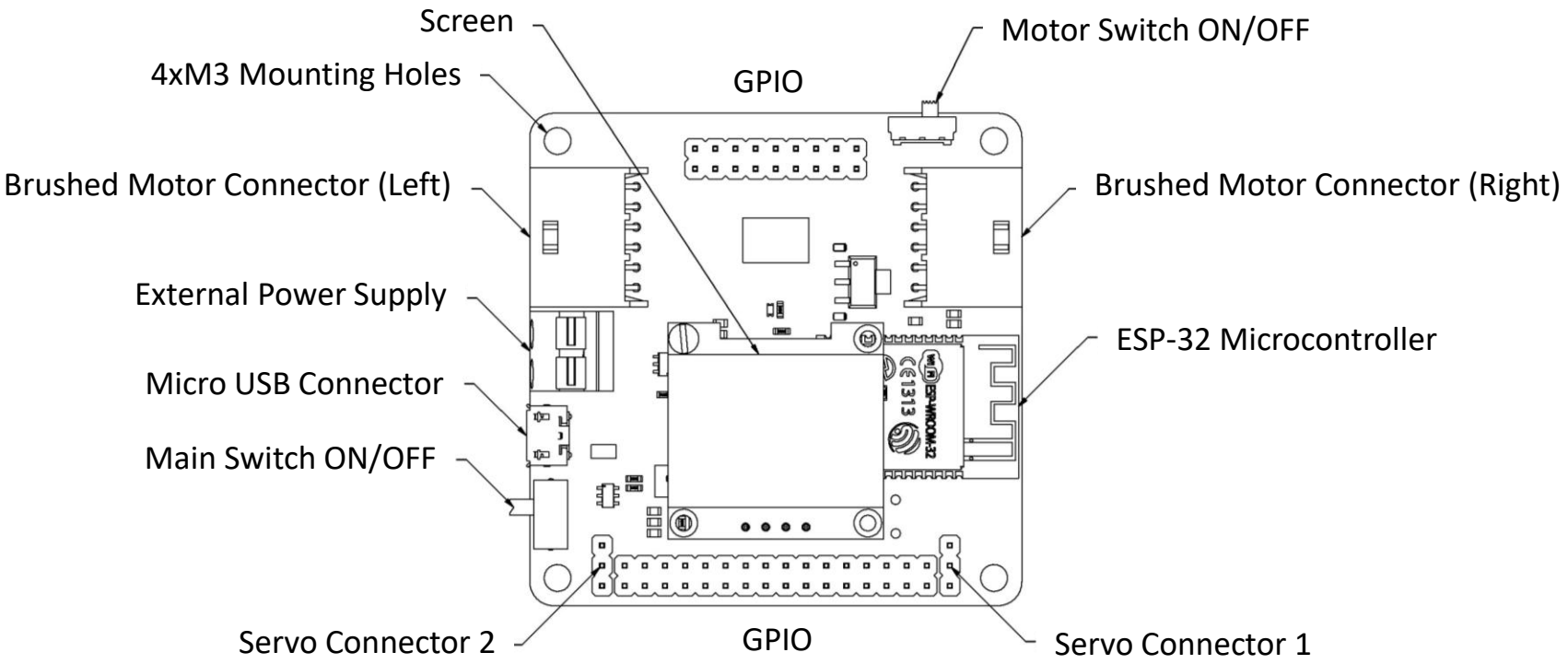
Instructions:

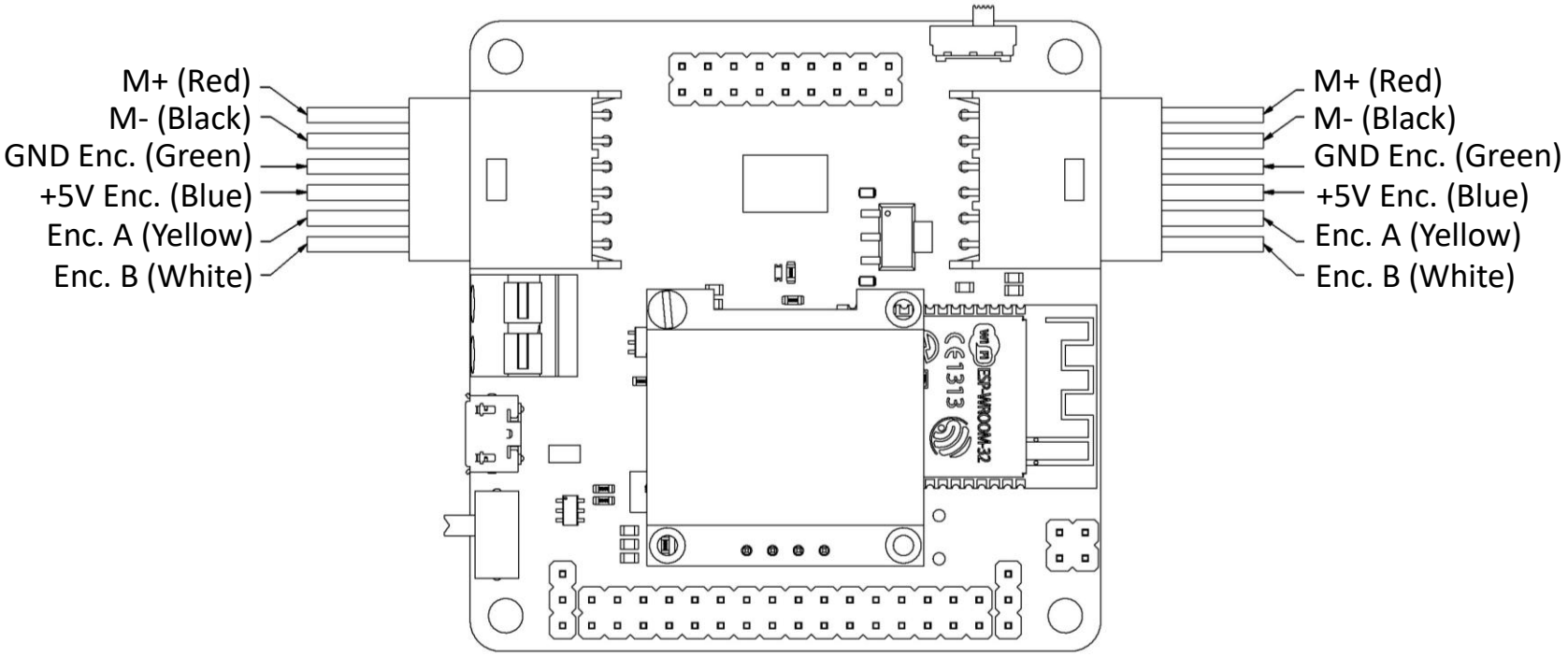
1.- Locate the mounting posts indicated in the diagram.

2.- Align the frame with the mounting posts.

3.- Slide the frame forward until it clicks firmly into place.

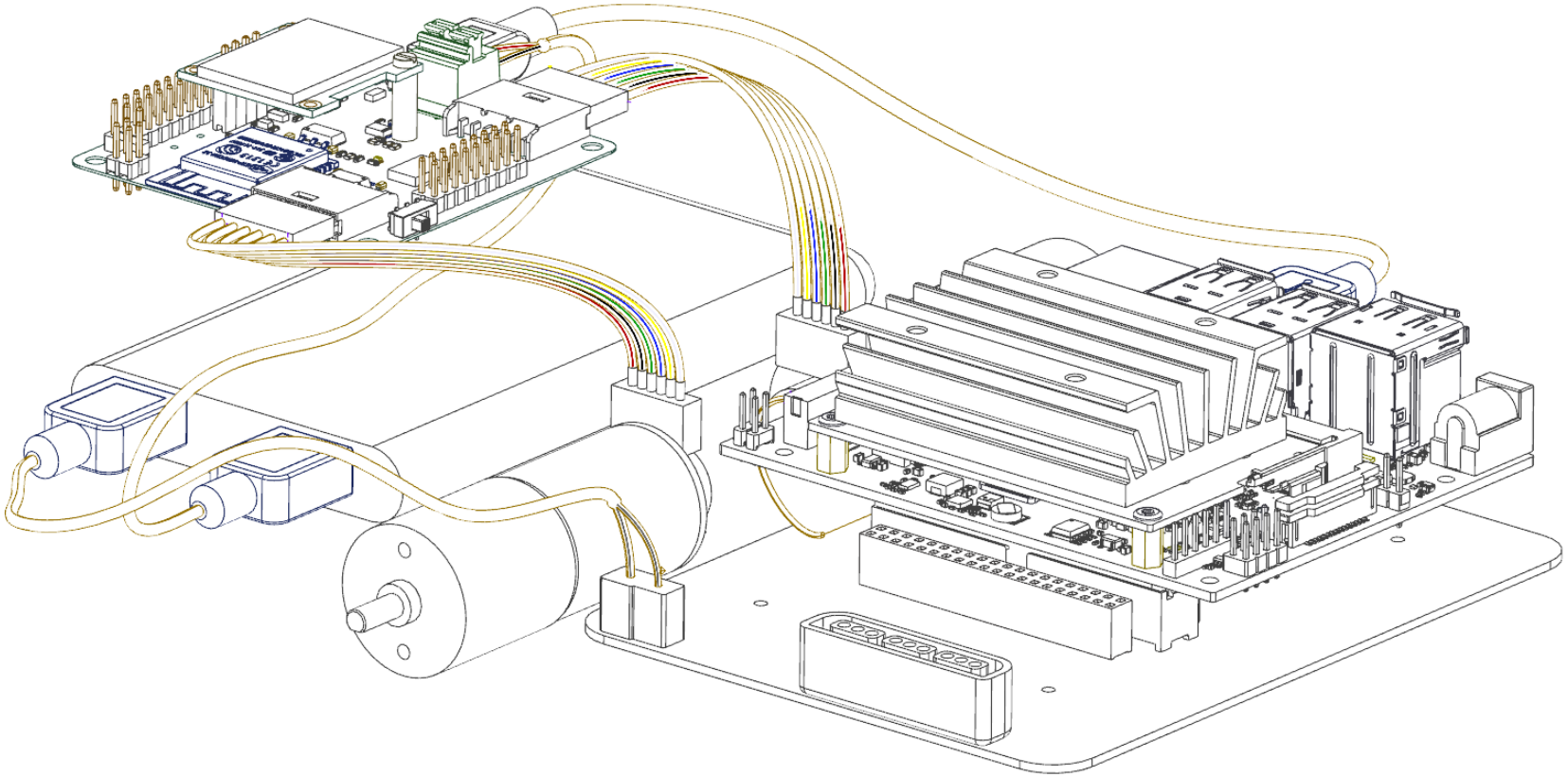
Warning: Do not force the component. Verify that all four tabs are properly seated in the mounting posts.


Screen
4xM3 Mounting Holes
Brushed Motor Connector (Left)
External Power Supply
Micro USB Connector
Main Switch ON/OFF
Servo Connector 2
GPIO
Motor Switch ON/OFF
GPIO
Brushed Motor Connector (Right)
ESP-32 Microcontroller
Servo Connector 1



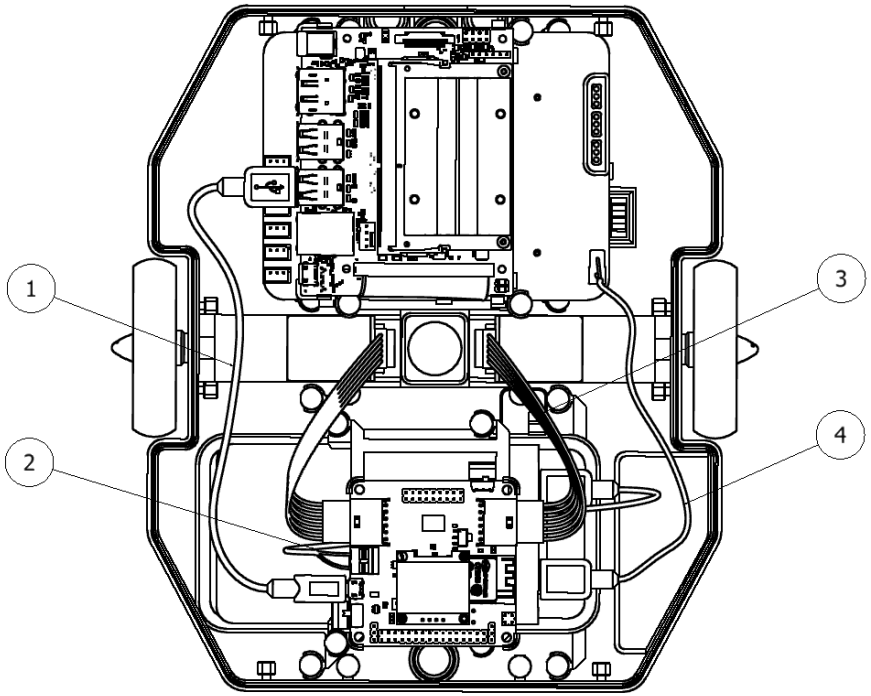
Hackerboard

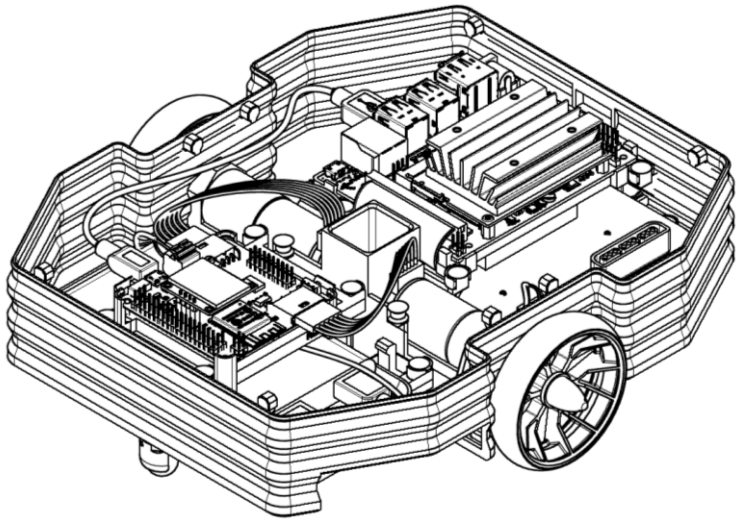
Motor Connection Diagram



18 Electrical Diagram

PARTS LIST		
ITEM	QTY	DESCRIPTION
1	1	USB - Micro USB Cable
2	1	USB Energy Cable
3	2	Motor Cables
4	1	USB Energy Cable with T connector





Wired Assembly

Instructions:

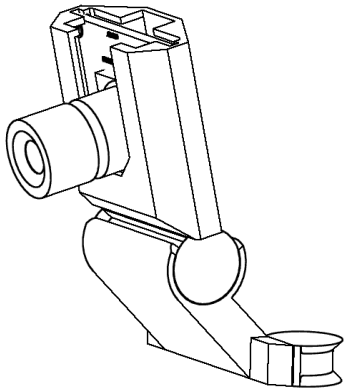
1.- Connect both motor cables from the DC motors to their corresponding terminal headers on the Hackerboard. Use **Figure 18.**

2.- Plug the USB to Micro-USB cable to connect the Jetson Nano with the Hackerboard.

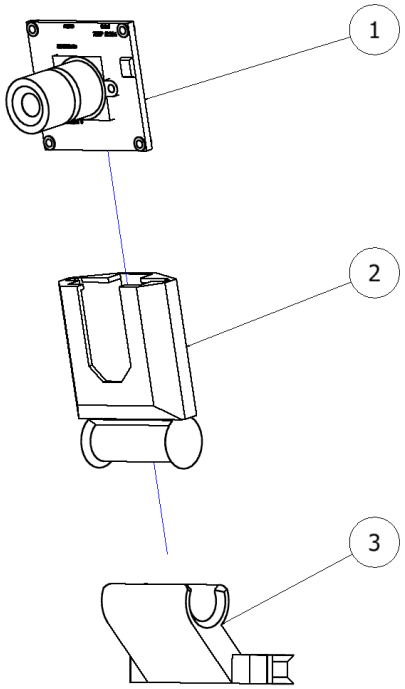
3.- Connect the USB energy cable in the Hackerboard's External Power Supply as shown in **Figure 17.**

Warning: Route all cables along the chassis perimeter channels, ensuring no loose wires interfere with the wheels or rotating assemblies.

PARTS LIST		
ITEM	QTY	DESCRIPTION
1	1	USB Camera Module
2	1	Camera Holder
3	1	Camera Holder Mount



Camera Holder Assembly



Instructions:

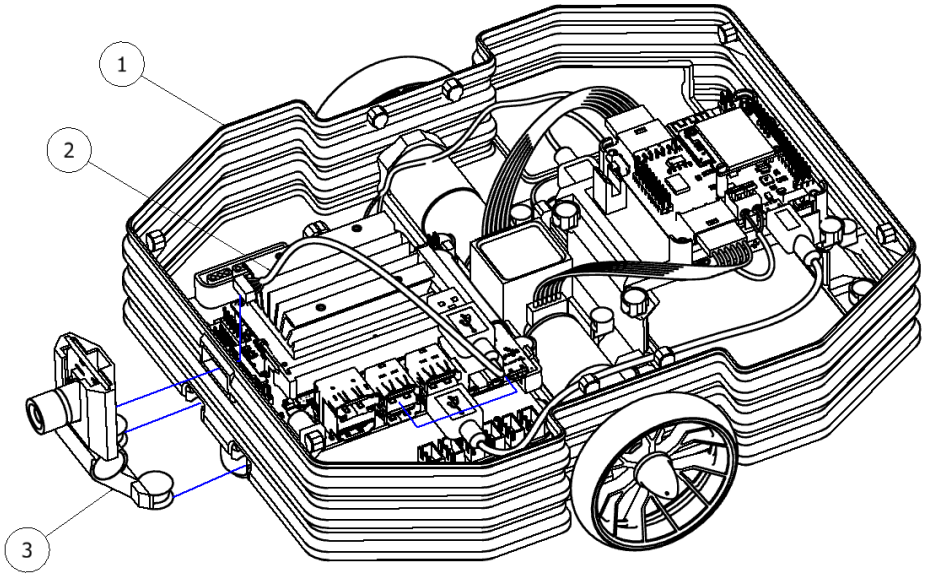
1.- Snap the Camera Holder onto the Camera Holder Mount until it firmly clicks into place.

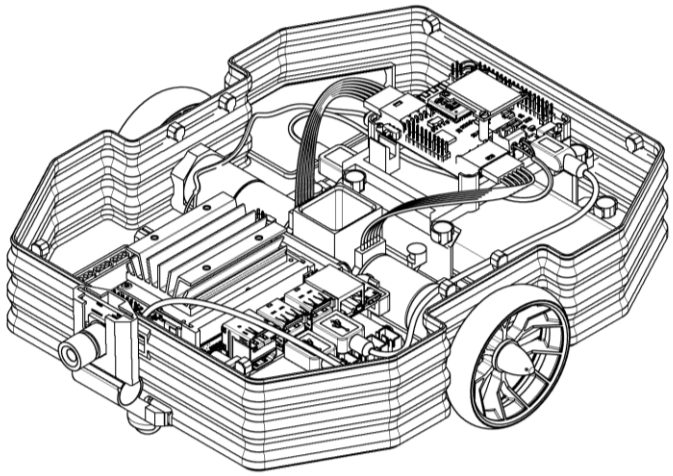
2.- Orient the Camera Module so that the cable connector is located at the bottom.

3.- Slide the camera downward into the Camera Holder from the top until it is seated.

Warning: Verify that the connector is facing downward before inserting the camera.

PARTS LIST	
ITEM	DESCRIPTION
1	Wired Assembly
2	USB Camera Cable
3	Camera Holder Assembly





Camera Assembly

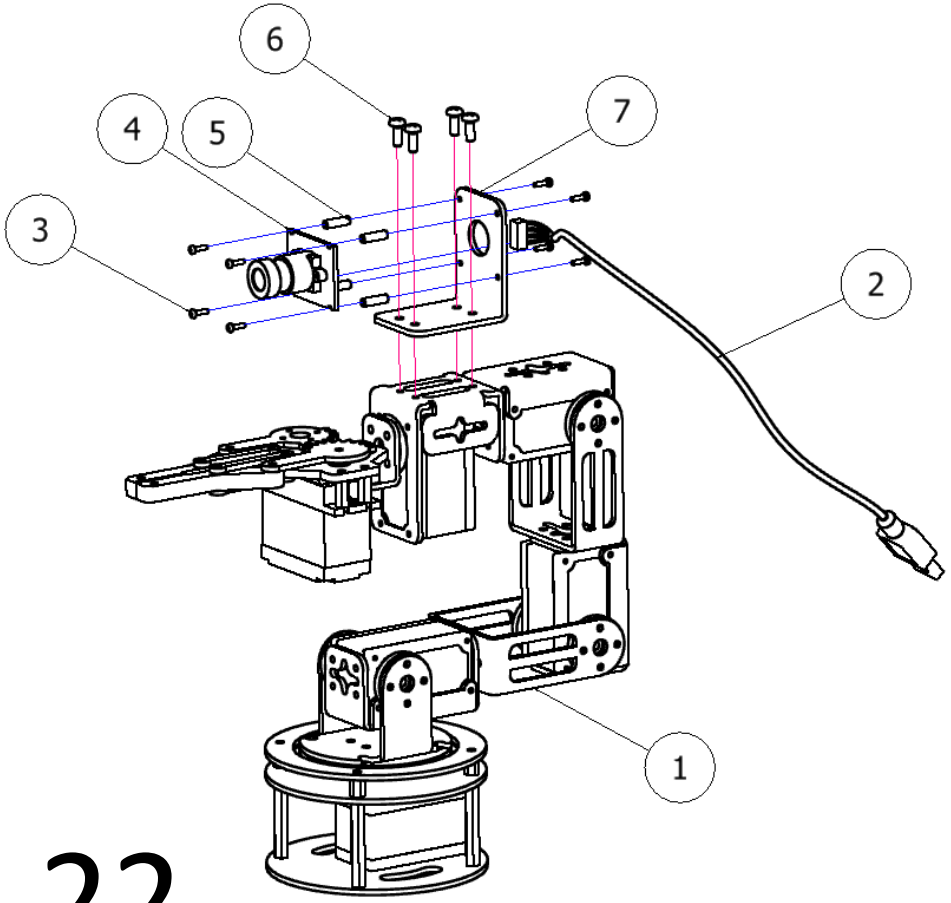
Instructions:

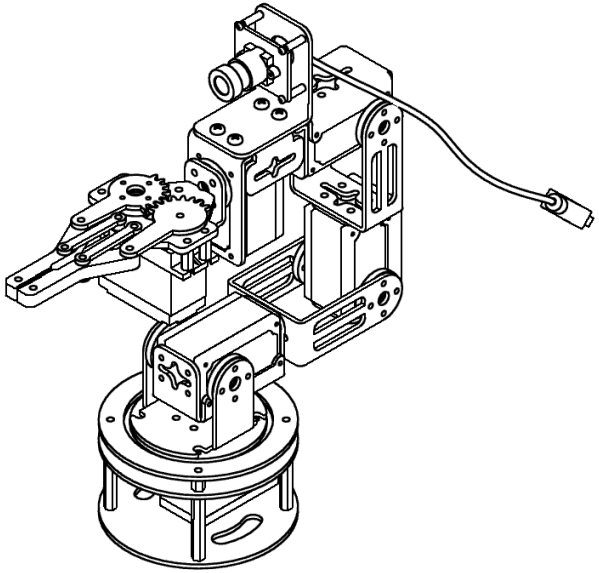
1.- Pass the USB Camera Cable from inside the chassis through the opening in the front of the base.

2.- Plug the USB Camera Cable into the camera connector located at the bottom of the camera module.

3.- Align and mount the camera holder assembly onto the front mounting slots of the chassis until firmly seated.

PARTS LIST		
ITEM	QTY	DESCRIPTION
1	1	Yahboom Robotic Arm
2	1	USB Camera Cable
3	8	M1.6x6 mm Pozi Pan Head Screw
4	1	USB Arm Camera Module
5	4	Spacer
6	4	M3x8 mm Pozi Pan Head Screw
7	1	Arm Camera Bracket





Arm Camera Assembly

Instructions:

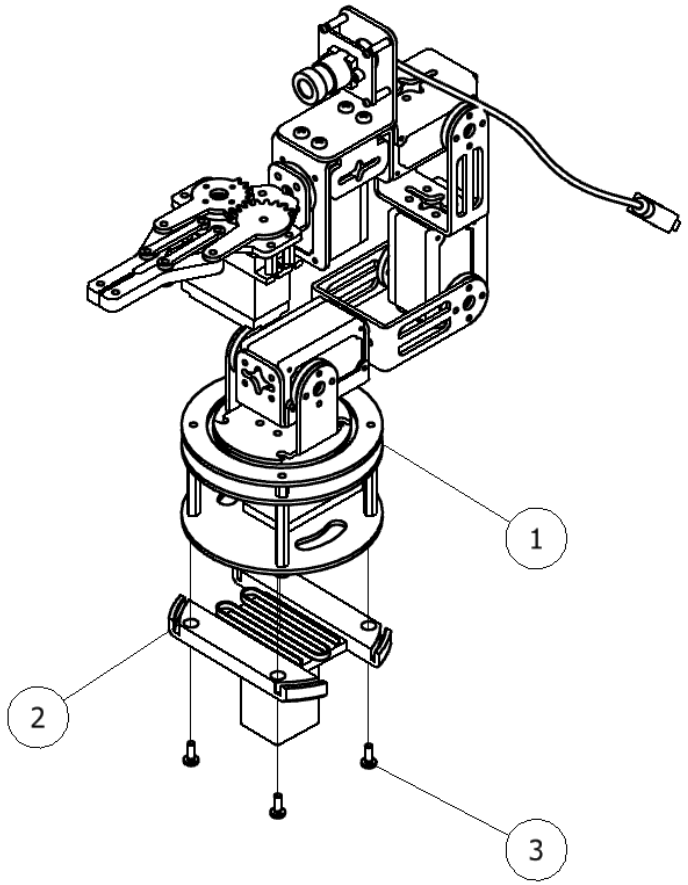
1.- Mount the Arm Camera Bracket using the screws as shown in the figure.

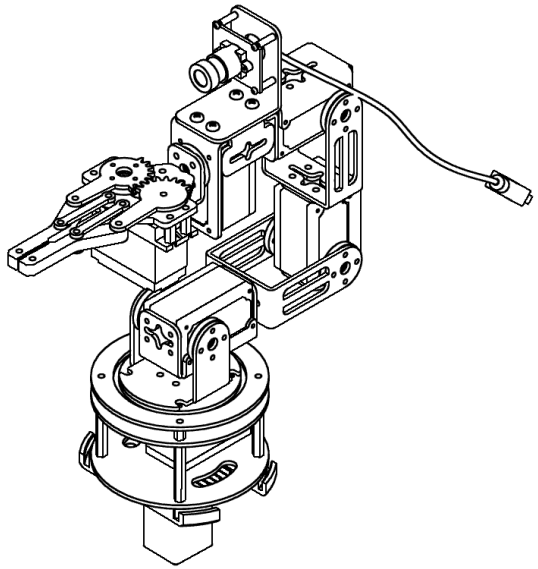
2.-Insert the USB Camera Cable through the center cutout of the arm camera bracket and connect it to the back of the USB Arm Camera Module.

3.- Use the screwdriver to properly tigthen the screws as shown in the figure.

<u>Warning:</u> Do not use excessive force.

PARTS LIST		
ITEM	QTY	DESCRIPTION
1	1	Arm Camera Assembly
2	1	Arm Bracket
3	4	M3x8 mm Pozi Pan Head Screw





Arm Base Assembly

Instructions:

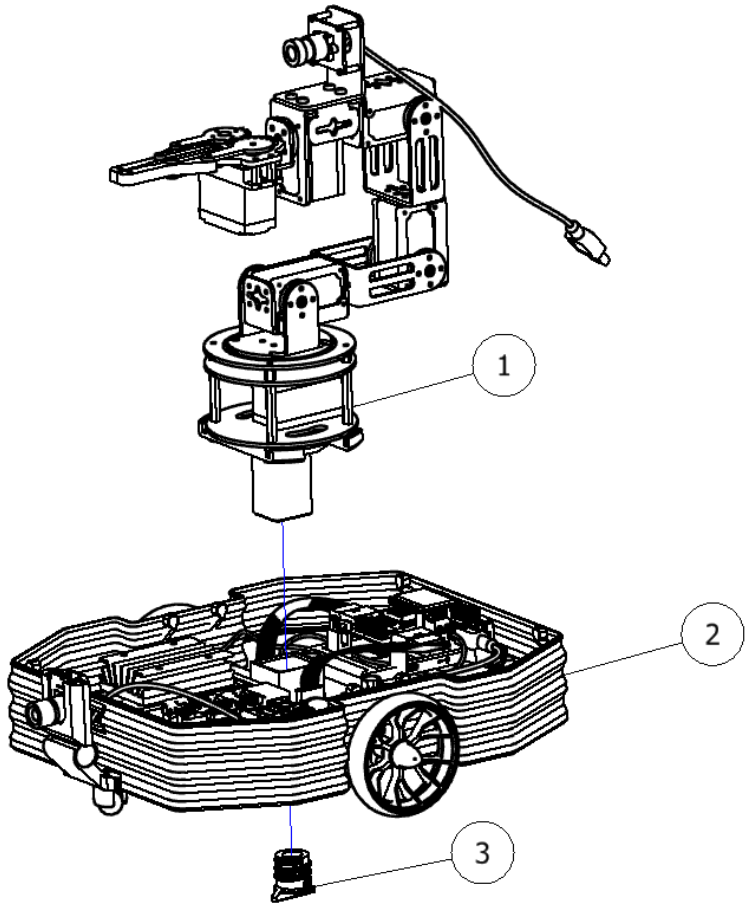
1.- Align the Arm Camera Assembly with the Arm Bracket, matching the four mounting holes.

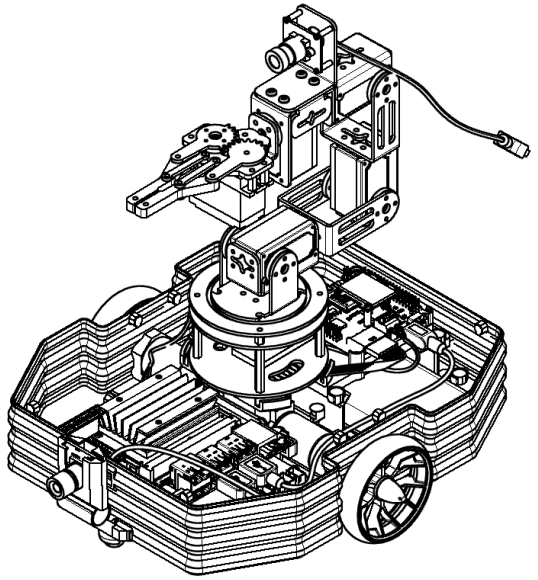
2.- Secure the Arm Bracket from underneath using the M3x8 mm Screws.

3.- Tighten the screws.

Warning: Do not use excessive force.

PARTS LIST	
ITEM	DESCRIPTION
1	Arm Base Assembly
2	Camera Assembly
3	Plastic Thumb Screw





Arm Chassis Assembly

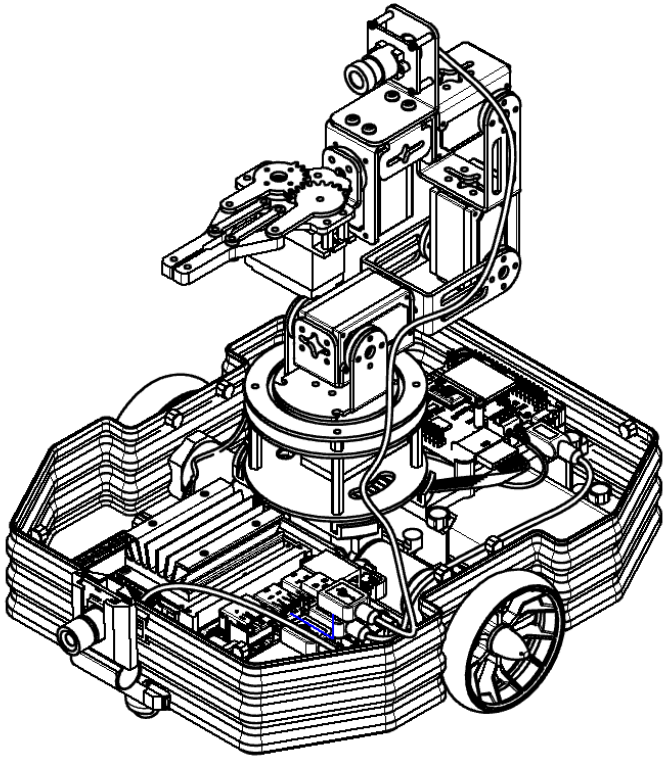
Instructions:

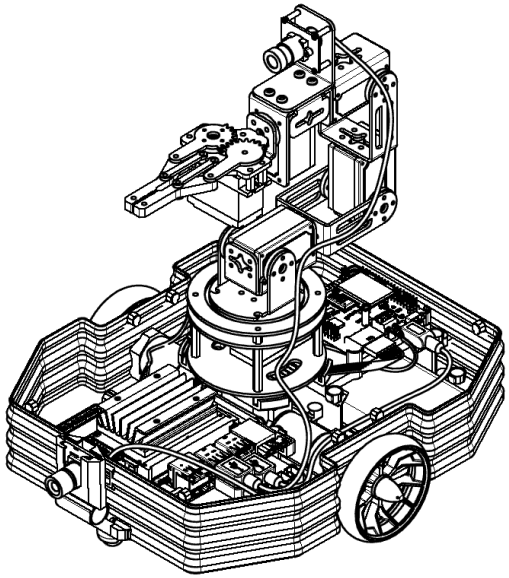
1.- Lower the Arm Base Assembly onto the top mounting opening of the chassis.

2.- From beneath the chassis base, insert the Plastic Thumb Screw into the center mounting thread.

3.- Hand-tighten the Plastic Thumb Screw clockwise until the Arm Assembly is firmly secured in position.

Warning: Do not use excessive force. Check the orientation of the arm before tightening.

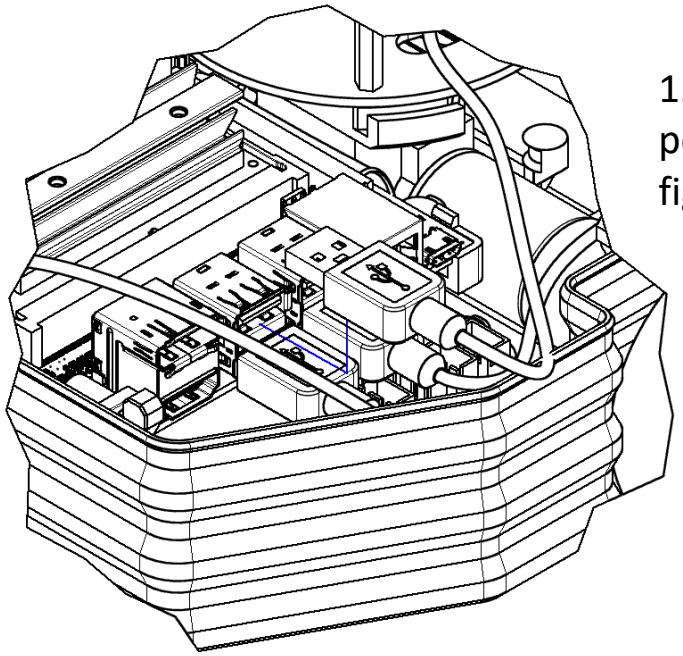




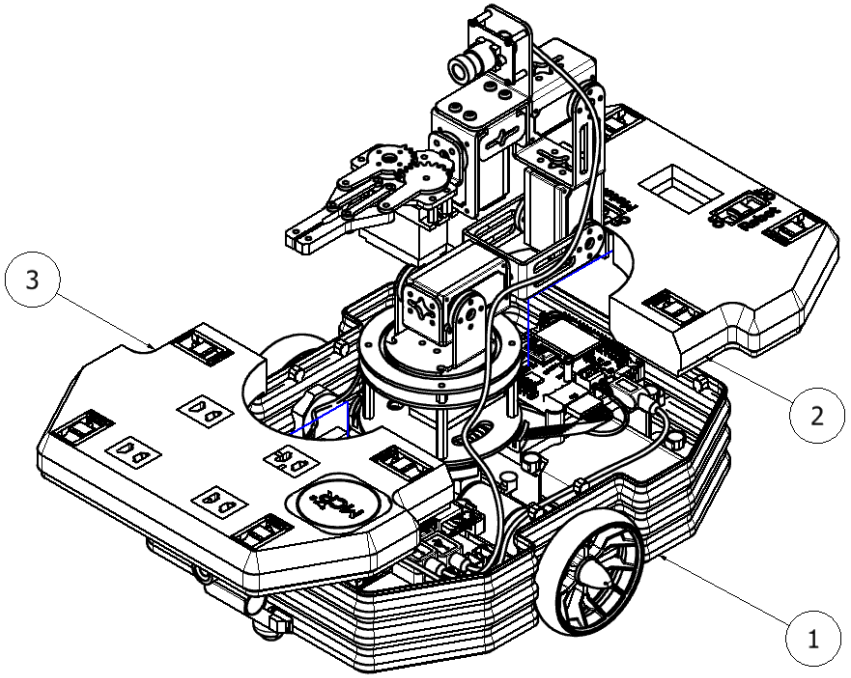
Arm Camera Connection Assembly

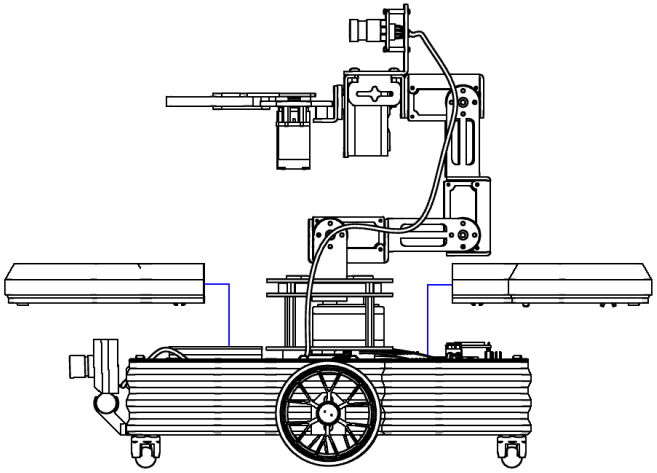
Instructions:

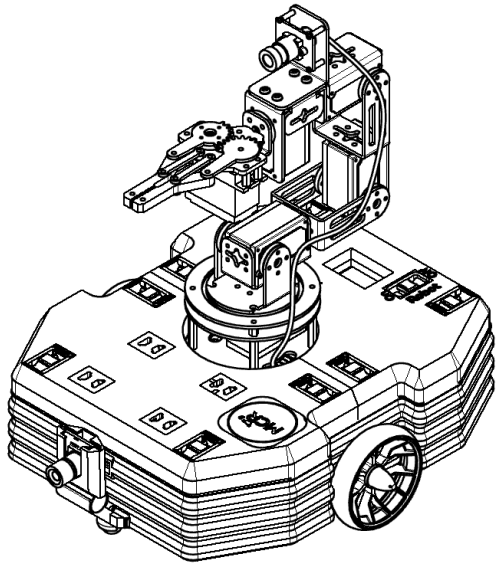
1.- Connect the USB Camera Cable to the port in the Jeston Nano as shown in the figure.



PARTS LIST	
ITEM	DESCRIPTION
1	Arm Camera Connection Assembly
2	Rear Top Cover
3	Front Top Cover







Top Cover Assembly

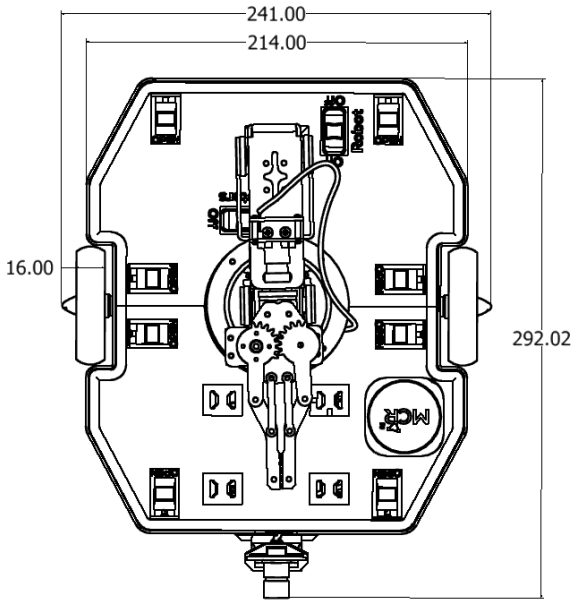
Instructions:

1.- Verify that all slide lock buttons on both Top Covers are set to the OPEN position prior to installation.

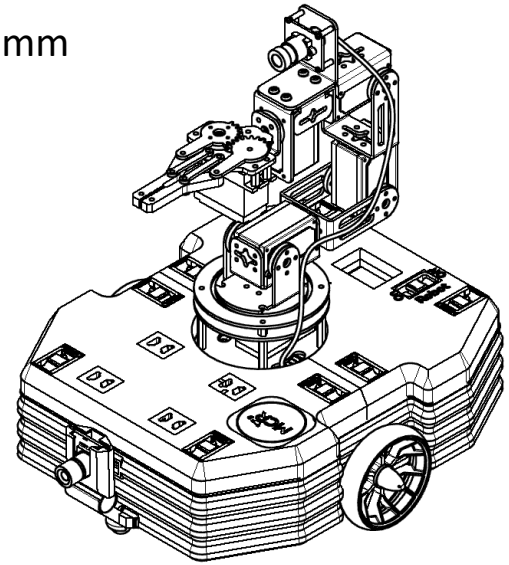
2.- Lower the Top Covers onto the chassis, aligning its cutout around the arm turntable base until it seats completely flat.

3.- Slide all lock buttons to the CLOSE position to secure both covers firmly to the chassis.

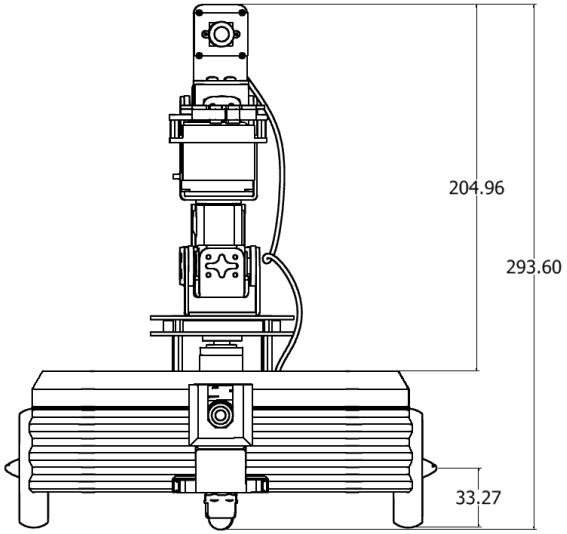
Warning: Do not force the covers down if they do not seat flat. Verify that internal wiring is completely tucked into the chassis channels.

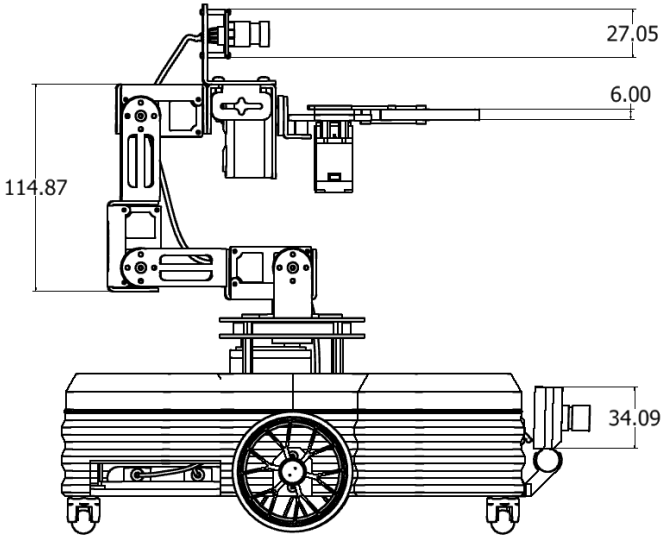


*All dimensions are in mm



PuzzleArm





General Dimensions